

OASIS — Oil & Asset Surveillance Intelligence System

Field Operations Specification

A buildable specification for real-time monitoring, anomaly detection, and intervention alerting across large multi-well fields under harsh and degraded conditions. Every rule carries a defined number or an explicit "supply per site / unarmed until provided" tag. Model-based capabilities that need data or infrastructure not present at first deployment are in Section 16, each with the condition that turns it on; they are not part of this release.

Effective date: immediate. Owner: Operations Technology. Review: quarterly + post-incident.

1. Mission & operating rules

OASIS watches up to ~4,000 wells at a 3–5 s sample rate and delivers real-time monitoring, anomaly detection, and intervention alerts under degraded conditions: sensor loss, comms cuts, heat, dust, and interference.

All timing rules are **wall-clock time plus a minimum count of valid samples**, never sample count alone, because cadence varies. Fast safety channels run at sample rate; slow reservoir channels run on 1-minute averages.

Operating rules that drive the code:

1. Safety first; when unsure, take the safe action.
2. Never hide a degraded or missing sensor; surface it.
3. Every alert states its data sources, what crossed which limit, the well state at the time, the data-quality context, and a recommended action.

4. No hard trip on a single raw sample, except the H₂S rapid-escalation level (Section 6.2), which is median-filtered first.
 5. Every trigger is judged against the well's state first.
 6. Compare against two baselines — anchored (fixed) and rolling (recent). Never alarm off the rolling one alone.
 7. H₂S is never suppressed by any state, mode, grace window, profile, or calibration condition.
 8. Modes go VERIFIED → DEGRADED → SURVIVAL by `degradation_score` (Section 3.4); logic complexity drops with them (Section 12).
 9. Any score may only lower autonomy automatically; raising it needs a human.
 10. **Production-trend triggers use flowing-only data.** Any trigger that measures a slow production or reservoir trend (decline, water-cut trend, GOR drift) is computed only over samples in the FLOWING state. SHUT_IN, TESTING, STARTUP, and RAMPING samples are excluded from those averages and baselines, so planned non-production never reads as a reservoir event.
 11. **No fixed absolute threshold survives a baseline near it.** Any threshold expressed as a bare number is suspect on a well whose normal operating point approaches that number — it will alarm continuously. Every absolute threshold is reviewed against per-well baselines and resolved one of two ways: **non-safety thresholds become per-well or baseline-relative** (GOR, water cut, sand, vibration); **safety bands stay absolute** (H₂S, MAWP, MAASP); a well whose baseline approaches a safety band is **blocked at go-live until the measurement is moved to where people are** (breathing-zone detection, Section 6.2), never by lowering the band. This review is mandatory whenever a new well type is onboarded — and it covers instrument ranges too: a default `instrument_span` that fits a light-oil well will quarantine a thermal well's real high-temperature readings, so spans are set to each well's true operating envelope at commissioning.
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2. Telemetry contract

2.1 Live channels

- Identity/time: `timestamp`, `well_id`.
- Wellhead/integrity: `pressure_psi`, `annulus_a_pressure_psi`, `annulus_b_pressure_psi`, `annulus_c_pressure_psi`, `wellhead_temp_f`.
- Flow control: `choke_position_pct`, `line_pressure_psi`, `custody_flow` (allocation/custody-meter reading, for the theft balance arm).
- Production (virtual/derived): `oil_flow_bbl_day`, `gas_flow_mscf_day`, `water_cut_pct`, `gor`, `temperature_f`.
- Integrity/safety: `h2s_ppm`, `h2s_cal_age_days`, `sand_rate` (unit per detector — Section 2.4).
- H₂S redundancy: `h2s_ppm_b`, `h2s_ppm_c` — second/third heads where fitted, for the 2-of-3 vote. One-head wells use the single-head rules in 6.2.
- ESP: `esp_vibration`, `esp_motor_amps`, `esp_frequency_hz`, `esp_intake_pressure_psi`, `esp_motor_temp_f`.
- Environment: `weather_condition`, `wind_velocity_kmh`, `wind_direction_deg`, `visibility_m`, `ambient_temp_f`.
- State: `well_state`, `state_source`.
- Baseline/power: `baseline_age_days`, `power_quality_flag`.
- Data quality: `sensor_health`, `gap_flag`, `gap_duration_s`, `comms_latency_s`, `degradation_score` (computed — Section 3.4).
- Provenance: per-sample signed hash where signing exists; until then a `provenance_ok` flag handled per 4.3.

2.2 Per-well reference data (supplied at commissioning; carried with an age and owner)

`mawp_psi`, `maasp_psi`, `esp_flg_amps`, `esp_freq_setpoint_hz`, `gas_handling_limit`,
`wcut_limit_pct` (default 70), `decline_forecast`, `hydrate_curve` / `wax_point_f`,
`pipe_geometry`, `instrument_span` per channel, `slew_max` per channel,
`h2s_sour_baseline_ppm` (the well's normal ambient H₂S, for the relative notify in 6.2),
`esp_vib_ref` (a clean vibration baseline frozen from a known-good window, for 6.5).

2.3 Site reference data

Muster-zone map (occupied-area coordinates), false-positive cost table per event type, alarm priority + max-response-time table.

2.4 Sand units

Sand is reported as `sand_rate` in the unit of the installed detector (acoustic counts or g/s; intrusive/ER probe metal loss; surface sampling in pounds-per-thousand-barrels). "ppm of sand" is not a valid field unit. Sand thresholds (6.7) are set per detector type at commissioning; there is no universal sand number.

Any trigger whose reference field is missing declares itself **UNARMED + reason** and never silently passes.

3. Data pipeline survivability

3.1 Layer-0 validation (every sample, before any trigger)

Each channel is despiked with a 5-sample median (Hampel) filter. A sample is VALID only if it passes all four checks; failures quarantine to proxies.

- **Range:** within `instrument_span`. Defaults: pressure/annulus/intake 0-5000 psi, H₂S 0-100 ppm, vibration 0-50 mm/s, temp 0-400 °F, WCUT 0-100 %, GOR 0-10000

scf/bbl, amps 0–300 A, freq 0–70 Hz, wind 0–150 km/h. → fail = SUSPECT. **The default temperature span (0–400 °F) is unsafe on thermal-recovery wells, where wellhead temperature legitimately exceeds it during steam breakthrough.** On any thermal/steamflood well the temperature (`instrument_span`) must be raised at commissioning (e.g. 0–600 °F) — this is a **go-live gate**: until the span is set for the well's real operating range, a genuine high-temperature event would be quarantined as a sensor fault and lost. A range check must never silently discard a reading the well is physically capable of producing.

- **Slew (physical-rate ceiling; faster = sensor glitch):** pressure 200 psi/s, annulus 100 psi/s, temp 5 °F/s, WCUT 1 %/s, H₂S 10 ppm/s, vibration 5 mm/s per s, amps 50 A/s. → fail = spike rejected. **The slew check is suspended for a channel during, and ±30 s around, a registered well-state change** (startup, shut-in, ramp, test), because legitimate transition steps are not glitches.
- **Frozen:** same value repeated ≥ 15 samples on a live channel. → fail = FAILED (flatline).
- **Staleness:** no new sample for > 2× the rolling-median interval for that well (floor 10 s). → STALE.

3.2 H₂S sensor honesty

(`h2s_cal_age_days`) past the calibration interval (default 30 days; 14 days under Extreme Heat, where electrochemical cells drift faster) caps confidence and raises a work order. It never enables suppression. A stale or failed gas sensor reading zero is not evidence of no gas.

3.3 Gap handling (split ladders)

Safety channels (H₂S, wellhead & annulus pressure): ≤30 s interpolate; 30–60 s caution + proxy; 60–300 s alarm (channel blind) + proxy + escalate; >300 s worst-case posture via ring escalation — gas present until proven otherwise; never an automatic field shutdown.

Production/reservoir channels: longer ladders; excluded from baselines past 5400 s of gap.

3.4 **degradation_score** (computed, 0-100)

Per console/cluster each cycle, the **maximum** of these sub-scores (worst dimension wins):

- **Sensor** = $40 \times \text{fraction FAILED} + 20 \times \text{fraction SUSPECT} + 10 \times \text{fraction DRIFT_SUSPECT}$ (cap 100).
- **Comms** = NOMINAL 0 / DEGRADED 15 / IMPAIRED 40 / PARTITIONED 70 / ISOLATED 100.
- **Gap** = 0 / 40 (safety channel blind 60-300 s) / 80 (>300 s).
- **Provenance** = 60 if any provenance break is active, else 0.

Weights are defaults to confirm in tuning; the formula is fixed.

3.5 Mode mapping (hysteresis)

0-33 VERIFIED. 34-66 DEGRADED (widen T2/T3 margins ~15%, human veto on T2+, ensemble required, Simplify stage 1). 67-100 SURVIVAL (T1 hard only, optimization suppressed, SAFE CORE). Upgrade needs ≥ 5 points into the better band for ≥ 15 min; downgrade is immediate.

3.6 Sensor health classes (defined tolerances)

VERIFIED = independent sources agree within 5% of span. PLAUSIBLE = single consistent source. DRIFT_SUSPECT = 4-8% deviation. SUSPECT = contradicts another source by >15% of span. FAILED = flatline/gap. GHOST = known bad tag. INTERPOLATED = ensemble-filled. H₂S is exempt from gating for alarm purposes; for the H₂S channel, a FAILED reading means "gas possible — alarm and verify," not "ignore."

3.7 Comms latency bands

<30 s NOMINAL · 30–90 DEGRADED · 90–300 IMPAIRED (suspend non-safety) · 300–1800 PARTITIONED (SURVIVAL) · >1800 ISOLATED. Heartbeat timeout 60 s; lost after 90 s; local SURVIVAL after 4 h partitioned.

4. Well-state machine & state trust

4.1 States

FLOWING, SHUT_IN, STARTUP, RAMPING, TESTING, WORKOVER, UNKNOWN. UNKNOWN defaults to FLOWING set-points + a flag. STARTUP grace 60 min (life-safety still armed). RAMPING ± 10 min for registered moves only.

TESTING/WORKOVER: planned suppression with audit — H₂S and the kick precursor are never suppressed. Non-FLOWING states are excluded from all production-trend statistics (rule 1.10).

4.2 State probability (defined fusion)

P(state) is a weighted vote, normalized over the sources present:

- Real cryptographic signature: 0.6.
- Operator input (unsigned): 0.2.
- Physics pattern: 0.2 — off until the signature library exists (Section 16); until then renormalize over the others.
- Visual confirmation: corroborator, +0.1 when present.

4.3 Trust ladder

- Real cryptographic signature → full suppression rights.

- A plain "signed" flag with no real signature → provisional only, ≤ 10 min, treated as unsigned. A flag is text; it does not earn suppression.
- INFERRED → advisory conditioning only. UNKNOWN → FLOWING set-points (fail-armed).

Suppression needs $P(\text{suppressive state}) \geq 0.85$ **and** a real signature. Where real signing is not deployed, no state earns full suppression; the most a state buys is its registered grace window. A declared state that contradicts measured physics reverts instantly to FLOWING set-points (fail armed) and raises a security alert. H₂S is suppressed by nothing.

5. Alert architecture

5.1 Budgets & flood control

T1 unlimited; T2 ≤ 18 ; T3 ≤ 35 concurrent per console. Per EEMUA 191 / ISA-18.2: 3% span deadband + off-delay on every analog alarm; priority + max-response-time per alarm from the site table; first-10-min upset cap ~ 10 then flood suppression (first-out / root-cause only); chatter guard auto-shelves > 3 cycles/min; steady-state target < 1 alarm / 10 min / operator.

5.2 Grouping

Group by same well + same trigger code within a time window, and by same declared manifold within window W (default 60 s) for a supervisor roll-up. No causal claim and no graph layer in this release.

5.3 Tiered gas response

Single head HIGH (20 ppm), low rate, plume clear of zones → shelter-advise + verify. HIGH + rising rate or plume toward a zone → local muster. 2-of-3 HIGH or any HIGH-HIGH (50 ppm) → full muster + ESD per 6.2. IDLH (100 ppm) → full emergency. If the

muster-zone map is missing, default to muster all nearby zones on HIGH+. Every muster gets a post-event review; false musters are a first-class cost (Section 9).

5.4 Load handling

The Simplification Governor (Section 12) acts on a plain count: when decision-required items per console exceed a ceiling (default 25 open, or T2/T3 over budget), batch T3 and defer advisories. Detection and logging are never throttled. Flat cap of 20 active items per operator.

6. Triggers

Rules: check `well_state` first; gate by `sensor_health`; H₂S exempt from suppression. Fast channels at sample rate; slow channels on 1-min averages with CUSUM/EWMA using robust σ (MAD). Persistence = $\geq T$ seconds AND $\geq N$ valid samples. Rate of change = windowed regression slope, never sample-to-sample. Every comparison uses anchored AND rolling baselines. Production-trend arms use flowing-only data (rule 1.10).

Cold-start rule (every learned trigger): a well lacking the required history arms on a seed baseline from OEM data + 3–5 same-manifold offset wells, runs tighter backstops, and labels output `SEED` until the real baseline matures. **Learned baselines are frozen from a clean, known-good window, not a rolling window that the developing fault can poison.** A learned trigger is never left unarmed because a well is new.

6.1 T1 — immediate / safety-critical

Persistence ≥ 30 s AND ≥ 8 valid samples unless stated. HIGH-HIGH latched, manual reset.

6.2 T1-B — H₂S (never suppressed)

H₂S is monitored as **two separate questions**, because a single sensor reading cannot

answer both. They use different measurements, different thresholds, and different responses.

Part A — "Is the area dangerous right now?" (area safety · absolute · breathing-zone)

This is the life-safety answer. It is measured at the **breathing zone** — where people actually are — from the area/point detectors, not from a source detector at the wellhead.

- Absolute bands, instantaneous, despiked: HIGH 20 ppm, HIGH-HIGH 50 ppm, rapid-escalation 65 ppm, IDLH 100 ppm. **Never relative, never lowered, never suppressed** — 20 ppm of H₂S in the air people breathe is dangerous regardless of what the well normally produces.
- **ESD executive action** fires on any of: (a) 2-of-3 heads at HIGH-HIGH; (b) a single head held at HIGH-HIGH (≥ 50) for ≥ 5 s; (c) any single validated sample ≥ 65 ppm; (d) HIGH-HIGH + rate ≥ 5 ppm/s.
- **Any validated HIGH-HIGH sample musters immediately**, regardless of duration — people are protected the moment gas is seen. Auto-ESD (which shuts in the well/area) waits for an executive arm above, because a nuisance ESD on a sub-5-second blip carries its own hazard (shut-in, restart, flaring). No HIGH-HIGH is ever silently dropped; it always musters.
- **Stale or single sensor:** a sustained HIGH-HIGH is still actionable — muster and escalate verification; do not wait for corroboration that cannot come.
- **Wind:** every gas alert carries plume direction; muster up-crosswind. At zero wind, assume omnidirectional and muster all nearby zones.

Part B — "Is something changing?" (change detection · baseline-relative) This is a separate, lower-priority question: is the well souring or is a condition shifting — independent of whether the area is dangerous yet. It is measured as a **rise relative to the well's** h2s_sour_baseline_ppm, on any detector.

- Notify when the **15-minute TWA rises ≥ 5 ppm above the well's sour baseline**; also on rate ≥ 5 ppm/s, or a sustained upward trend.

- For a sweet well with no sour baseline, this falls back to a 15-minute TWA ≥ 10 ppm.
- This arm fires on a *change*, not on a steady high baseline, so a chronically sour well does not flood it — and it catches a souring trend while the absolute level is still below HIGH.

Why the split: a chronically sour wellhead conflates the two questions if read by one sensor — its steady ~18 ppm source reading looks like a permanent "area dangerous" alarm when the real situation is "the source is sour, the breathing zone may be fine, and nothing is changing." Part A answers danger from the breathing zone; Part B answers change from the baseline. Neither floods the other.

Commissioning gate (chronic sour baseline) — the 100% fix, by construction: a well whose `h2s_sour_baseline_ppm` is within 5 ppm of a safety band **may not go live until breathing-zone detection is engineered and verified**, so that Part A reads the concentration where people are (far below the source baseline with correct detector siting and ventilation). With breathing-zone detection in place, the absolute band is nowhere near the source baseline and the flood is impossible — not mitigated, eliminated. Until that detection is verified, the well is held **NOT-CLEARED for live operation**. The band is never lowered to fit the baseline; the measurement is moved to where the hazard to people actually is. This is a site-engineering requirement (a sensor and a layout), enforced as a go-live blocker rather than worked around in software.

Off-delay: below LOW for 60 s.

6.3 T1-A — well & casing integrity

WHP HIGH = +15% vs anchored. HIGH-HIGH = `mawp_psi` (unarmed if absent). LOW-LOW = -20% vs anchored or below `line_pressure_psi` (containment loss). SCP: thermally compensated annulus vs `maasp_psi`; with <24 h history, run uncompensated with a +10% wider margin and label `SEEDDED`. Compensated exceedance ≥ 30 min → integrity alert. **Thermal compensation must match the well's thermal driver.** On a normal well the compensation subtracts the learned **daily** thermal curve. On a **thermal-recovery (steamflood / cyclic-steam) well**, annulus pressure also follows the multi-day

steam-injection cycle, which a daily curve does not remove — so the compensation must additionally subtract the learned steam-cycle component (keyed to the injection schedule or the wellhead-temperature cycle). Without this, a thermal well produces a recurring `SCP_SUSPECT` nuisance on every steam cycle: a build that simply tracks temperature is thermal/operational, not casing communication. The diagnosis still holds — a build that *follows temperature* is thermal; a build that *persists when temperature falls* is communication — but the compensation must know about the steam cycle so it stops flagging the thermal one. **SCP status:** passive telemetry raises `SCP_SUSPECT` only. `SCP_CONFIRMED` is reserved for when a bleed-down/build-up test is actually run (pressure rebuild = communication). OASIS prescribes the test; it does not claim "confirmed" from passive data.

6.4 T1-C — catastrophic ΔP / kick precursor

Rate (30 s regression confirmed over 60 s): rising ≥ 50 psi/min = influx; falling ≥ 100 psi/min = leak. Suppressed in registered RAMPING ± 10 min; advisory in STARTUP grace; **never suppressed by TESTING**. SHUT_IN build judged vs historical signature (seed if new). During a storm, if the rate arm doesn't trip but pressure is sustained-rising, raise an advisory and confirm on a single trusted channel + physics rather than waiting for a second channel the storm has degraded.

6.5 T1-D — ESP failure

Vibration — three tiers off the frozen clean baseline `esp_vib_ref`:

- **Creep advisory:** flowing vibration **held $\geq 30\%$ above `esp_vib_ref` continuously for ≥ 6 h** — catches gradual wear that never reaches the ISO limit. The ≥ 6 h continuous hold is mandatory, not optional: it is what separates real wear (sustained for days) from normal diurnal peaks that briefly cross the level and fall back. Without the hold this arm false-alarms on healthy wells.
- **Alert:** +100% over `esp_vib_ref`, or crossing the ISO severity backstop (7.1 / 11 mm/s RMS, ISO 20816 class), whichever comes first. The clean baseline is frozen

from a known-good window, never a rolling median that the developing fault pulls upward. Rising motor temp is a supporting signal, not required — a vibration excursion alone alarms. **VSD bands:** underload $< 70\%$ $(\text{esp_fla_amps}) \geq 60 \text{ s}$; overload $> 110\%$ $(\text{esp_fla_amps}) \geq 30 \text{ s}$. Unarmed if (esp_fla_amps) absent. **Drive fault:** $(\text{esp_frequency_hz}) > 2 \text{ Hz}$ off $(\text{esp_freq_setpoint_hz})$ without a registered command. Unarmed if setpoint absent. **Gas-lock vs pump-off:** gas-lock = amps oscillation (rolling σ of amps $>$ threshold, default 15 A) + intake-pressure swing + flow drop. Pump-off = low steady amps + low intake + flow $< 60\%$ of 1-h flowing baseline. The amps-oscillation σ is the deciding signal. Flow is a supporting signal only (virtual). A minimal signature check stays alive even in SURVIVAL for the named failure modes. SHUT_IN expects zero flow/amps. Post-workover 30-day window runs tighter backstops, not looser.

6.6 T1-E — theft / interference (per regional profile)

Two arms, **both required for an executive theft alarm**, because either one alone is noisy: (a) a step drop on $(\text{line_pressure_psi})$ (or WHP) without a choke move — default step $\geq 10\%$ of line pressure within $< 120 \text{ s}$, persistence $\geq 120 \text{ s}$ + sister-line corroboration; (b) a **custody flow-balance mismatch** $\geq 5\%$ between (custody_flow) (allocation/custody-meter channel) and the wells' summed production. The pressure-step arm alone produces more false alarms than real ones (ordinary line-pressure noise and thermal dips cross the step threshold), so on its own it is **advisory only**. The executive alarm requires the balance-mismatch arm to agree. If no custody-meter channel exists, T1-E runs in **advisory-until-armed** mode — pressure-step advisories plus a daily balance reconciliation — and says so; it never raises an executive theft alarm on pressure alone.

6.7 T2 — slow / statistical (4-h cycle; informational, not latched; flowing-only)

Exclude FAILED/SUSPECT/gap and non-FLOWING samples from all means. Every T2 arm runs advisory-only until its window is filled and says so.

- **T2-A GOR:** CUSUM ≥ 1 robust- σ held $\geq 30 \text{ min}$ vs anchored + rolling; backstop = 4-h flowing median $\geq (\text{gas_handling_limit})$. No fixed number.

- **T2-B WCUT — three arms (per-well, not a fixed default):** (i) 4-h flowing median $\geq$ `wcut_limit_pct`, which **must be set per well** — a mature well that normally produces above a fixed 70% would otherwise alarm continuously. Where no per-well limit is supplied, the absolute arm fires on **the well's own water-cut baseline + 8 points** (baseline-relative), never a bare 70%, so a chronically high-water-cut well does not flood; (ii) fast step $\geq$ 15% of remaining oil fraction in 1 h; (iii) **slow-trend breakthrough:** a flowing water-cut regression slope $\geq$ **1.5 percentage points per week sustained over a $\geq$ 14-day window** vs the anchored baseline — catches a gradual breakthrough (a rise of ~10 points over a month) that never reaches the absolute limit and never steps fast. A 7-day window is too short for a breakthrough this slow; the slope is measured over 14 days as a regression. **For a mature, chronically high-water-cut well the trend arm is the primary signal** — the absolute arm tells you the well is wet, which you already know; the trend tells you it is getting worse.
- **T2-C Depletion (flowing-only):** 24-h flowing median $> 2\sigma$ below `decline_forecast`. With < 24 h flowing history or no forecast, advisory-only via BASELINE_WALK.
- **T2-D Sand:** rate step $\geq 2\times$ rolling baseline held ≥ 5 min; 1-h median and single-reading limits set per detector type (Section 2.4). Erosion budget per API RP 14E using `pipe_geometry` + velocity; unarmed if geometry absent.
- **T2-E Hydrate/Wax:** JT-cooled temp within 3–5 °F of `hydrate_curve` at line pressure; wax vs `wax_point_f`. Unarmed until the curve/point is supplied. Never uses flat wellhead temp alone.
- **T2-F Injector / T2-G Corrosion:** unarmed until their channels exist.

6.8 T3 — 24-h cycle (flowing-only)

T3-ESP: 24-h vibration mean $+20\%$ vs `esp_vib_ref`. T3-DECL: **24-h flowing flow $\leq 90\%$ of the 7-day flowing baseline** — both averages exclude SHUT_IN/TESTING/STARTUP/RAMPING, so planned non-production never trips a

decline alarm. Advisory-only until windows are filled. T3-CHEM/FLARE unarmed until channels exist.

6.9 Edge vs central split

Edge (survives comms loss): all Layer-0, all T1, gas response, ring escalation, mode logic, `degradation_score`. Central: T2/T3 statistics, cross-well roll-ups, reporting.

7. Baselines & re-anchoring

Anchored (engineering-signed; moves only by re-anchor protocol), intermediate (90-day flowing median), rolling (24-h flowing median). Clean reference baselines (`esp_vib_ref`, sour baseline) are frozen from known-good windows. Mandatory re-anchor on workover, stimulation, ESP change, confirmed character change, choke/tubing change; scheduled review at 12 months. `baseline_age_days` > 365 → BASELINE_STALE advisory; > 540 → dependent learned thresholds drop one confidence class. Stale baselines are surfaced loudly, and the confidence of dependent thresholds decays as age grows.

8. Confidence & explainability

- **Confidence is coarse, not calibrated.** Each alert carries a confidence from sensor health + how far past the limit + how many valid samples backed it, labeled `coarse`. Statistical calibration applies only to probabilistic models (Section 16).
- **Explanation is "what crossed which limit."** Each alert shows the driving channel(s), the value, the limit, and the anchored/rolling deltas.
- **Counterfactuals and what-if render only when a model that produces them exists** (Section 16); they are not attached to every alert.

9. Feedback, validation & false-positive economics

- **Alert lifecycle fields:** id, well, trigger code, tier, fired_at, severity, state (OPEN→ACK→SUPPRESSED/ESCALATED→CLEARED), ack, closure code, group id, well-state-at-fire, `confidence_coarse`.
- **Closure:** OASIS proposes a code; one-click accept; required for later learning.
- **Validation:** replay labeled datasets per trigger and per well-state — precision, recall, lead-time — plus adversarial replays (spoofed state, frozen value, replay, theft). No trigger arms live ESD before passing replay.
- **FP economics:** track T1/T2/T3 false-positive and false-muster costs against ground-truth events using the supplied cost table; report per well-state and per profile. Two named measures are tracked explicitly: the shut-in/test false-decline rate and the sour-well low-notify rate, since both are known flood risks.

10. Regional profiles & extreme conditions

Ring escalation: Well → Cluster → Field, each step needing independent corroboration (second detector or sister-well signal; drone confirmation is a Section 16 item). De-escalation reverses with hysteresis (step down after the corroborating signal clears for 5 min). One silent sensor is a reason to look, not to shut a field.

Profiles (one-click; auto-detect by `weather_condition` + sour flag):

- **GCC Desert:** Shamal mode (non-optical bias; comms latency thresholds ×2); Extreme Heat (> 45 °C: widen thermal envelopes; H₂S calibration interval shortened to 14 days); sandstorm posture (extend confirmation windows per 6.4); high-H₂S sour config with the relative/TWA notify of 6.2.

- **Africa Delta / onshore:** T1-E armed; `power_quality_flag` conditioning on ESP electrical triggers; corrosion profile; extended-outage ring tuning.
- **Africa offshore:** hydrate/wax cold profile; slugging-aware flow; logistics-aware sparing.

Profiles tune numbers; they never touch H₂S bands, safety limits, or prohibited behaviours.

High-tempo fallback and standing modes (Shamal, mass outage, talent shortage, extreme heat) are time-boxed, logged, and auto-reviewed at 14 days; safety, state trust, and ring escalation stay fully active.

11. Response format

```
[LEVEL] ALERT – [CODE] – [WELL/CLUSTER]
Well State: <state> (P=0.xx, source=SIGNED|UNSIGNED|INFERRED|UNKNOWN)
Mode: VERIFIED|DEGRADED|SURVIVAL | Degradation Score: XX/100
Logic Profile: FULL|SIMPLIFIED-1|SAFE-CORE
Data Quality: sensor_health=[..] comms_latency_s=[..] gap_duration_s=[..]
               h2s_cal_age_days=[..] baseline_age_days=[..]
provenance=OK|BROKEN
DRIVING CHANNELS: [channel + value + limit crossed; anchored vs rolling
delta]
SISTER WELLS: [isolated vs systemic – manifold roll-up]
CONFIDENCE: [coarse] – sensor health + margin + sample count
IMMEDIATE ACTIONS: [one-click options; for SCP: prescribe bleed-down/build-
up test]
WIND (if gas): [direction/velocity; muster guidance; all-zones if calm]
Alert Lifecycle: id=[..] state=[..] well_state_at_fire=[..]
```

12. SAFE CORE & dynamic simplification

Every rule is tagged CORE or EXTENDED.

- **SAFE CORE** (self-contained, edge, no model dependency): Layer-0, all T1, hard state-trust (fail-armed), gap ladder, ring escalation, tiered gas, EEMUA flood control, mode logic, `degradation_score`.
- **EXTENDED**: everything model-based (Section 16).

Simplification Governor stages, by a plain item count:

- **FULL** — nominal.
- **SIMPLIFIED-1** — on DEGRADED mode, or open decision-items over the ceiling (default >25), or high-tempo: batch T3, defer advisories, hold last-good.
- **SAFE-CORE** — on SURVIVAL, comms partition, security field event, or manual switch: only SAFE CORE runs; every suppressed layer logged for replay; nothing constitutional is ever pruned.

The active profile is printed on every alert.

13. KPIs & trust score

13.1 KPI set (measured indicators only)

- Detection lead time per trigger class (P50/P90).
- Alarm-system health per EEMUA (rate/operator, flood frequency, chattering, stale alarms).
- Safety leading indicators: gas-detection coverage uptime, time-blind on safety channels, calibration-overdue count, muster-drill performance.

- FP cost and false-muster count; the shut-in false-decline rate; the sour-well low-notify rate.

Every KPI carries its definition, data window, and exclusions inline. The full set is published together; a favorable subset is not. No "incident probability reduction" is claimed. (Forecast-based KPIs begin when their module is live, Section 16.)

13.2 Trust score

Per trigger / well / cluster, computed on a rolling window of closed (confirmed) alerts — not on the clock, because outcomes arrive only when alerts are cleared. It is precision × recall over that window, with window size and confirmation lag stated. It can only lower autonomy. Trust scores apply to system parts, never to human operators.

14. Prohibited behaviours

1. Never clear a T1 without protocol.
2. Never rely on a single source for a critical trigger (the H₂S sustained-level and rapid-escalation arms are the designed exceptions).
3. Never hide degradation or missing data.
4. Never exceed alert budgets without escalation.
5. Never suppress H₂S or integrity triggers under any state, mode, grace, profile, or calibration condition.
6. Never hard-trip on a single un-validated sample (the despiked H₂S rapid-escalation level is the one exception).
7. Never alarm a slow reservoir channel at sample rate.
8. Never compute rate sample-to-sample; always a windowed regression slope.
9. Never evaluate a trigger without well-state context; never let a grace window touch a life-safety channel; never let TESTING suppress the kick precursor.

10. **Never include non-FLOWING samples in a production-trend statistic** — planned shut-ins and tests must never read as decline, water-cut, or GOR events.
11. Never compare only against a rolling baseline; never build a learned baseline from a window that contains the developing fault.
12. Never treat a stale or failed gas sensor's silence as no gas.
13. **Never use a fixed H₂S low-notify on a sour well** — the notify tracks the well's own sour baseline (+5 ppm); a fixed 10 ppm is only for sweet wells. HIGH and above stay absolute.
14. Never use flat wellhead temperature as the hydrate input.
15. Never grant full suppression on a plain "signed" flag — only a real signature earns it.
16. Never leave a post-workover ESP on relaxed thresholds.
17. Never silently disarm a trigger when its input goes suspect — degrade through substitutes visibly, and say a reference field is missing rather than passing silently.
18. Never escalate a single-well worst-case posture beyond its ring without independent corroboration.
19. Never present coarse confidence as calibrated; never claim "SCP confirmed" from passive data.
20. Never act on a value whose provenance is broken, except to alarm on it.
21. Never set a universal sand number — sand thresholds are per detector type.
22. **Never use a fixed absolute water-cut limit on a mature well** — the limit is per-well or baseline-relative; the rising-trend arm is the real signal on chronically wet wells.
23. **Never raise an executive theft alarm on the pressure-step arm alone** — it requires the custody balance-mismatch arm; pressure-step by itself is advisory.
24. **Never bring a chronic-sour-baseline well live without verified breathing-zone detection** (Section 6.2 gate); never lower a safety band to fit a baseline. Answer "is the area dangerous" (absolute, breathing-zone) and "is something changing" (baseline-relative) as two separate streams, never one conflated threshold.

25. **Never ship a new well type without the absolute-threshold review** (rule 11) — confirm no fixed threshold sits near that well's baseline before arming.
 26. **Never let a default instrument range discard a reading the well can physically produce** — thermal wells need a raised temperature span at go-live (Section 3.1), or real steam events are lost as sensor faults.
 27. **Never raise SCP_SUSPECT on a thermal build that tracks the steam cycle** — thermal-recovery wells compensate the steam-cycle component, not just the daily curve (Section 6.3).
 28. Never reject a legitimate state-transition step as a Layer-0 spike — the slew check is suspended around registered state changes.
 29. Never report a "safety incident probability reduction"; never report a favorable KPI subset.
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15. Threshold basis (current standards)

Threshold	Value	Basis
H ₂ S HIGH / HIGH-HIGH / rapid / IDLH	20 / 50 / 65 / 100 ppm	OSHA ceiling/peak; NIOSH IDLH 100; rapid between HH and IDLH. Safety bands stay absolute and read the breathing zone; a chronic-baseline well is blocked at go-live until breathing-zone detection is verified (6.2), never a lowered band
H ₂ S low notify	sour baseline +5 ppm (15-min TWA); 10 ppm only for sweet wells	Tracks the well's own ambient so chronic sour fields don't flood

Threshold	Value	Basis
H ₂ S ESD sustained-level	≥ 5 s at HIGH-HIGH	Long enough to reject a single despiked transient, short enough to catch a brief real spike
ESP vibration creep / alert	+30% held ≥6 h / +100% or ISO 7.1-11 mm/s	ISO 20816 severity; creep tier (with mandatory hold) catches sub-ISO gradual wear
VSD under/overload	70% / 110% FLA	Standard motor protection
WHP HIGH / HIGH-HIGH / LOW-LOW	+15% / MAWP / -20%	Operational deviation vs design limit
Kick / leak rate	≥50 / ≥100 psi/min	Surface-pressure heuristic; tune per well
WCUT arms	per-well limit (or baseline +8 pts) / 15%/h step / ≥1.5 pts/wk over ≥14 d	Per-well absolute (never a fixed 70 that floods mature wells) + fast + slow-slope; trend is primary on high-water-cut wells
GOR / sand	per-well learned / per- detector	Well- and instrument-specific; no universal number
Alarm discipline	3% deadband, ~10 first- 10-min, >3/min chatter	EEMUA 191 / ISA-18.2
Comms / heartbeat	30/90/300/1800 s; 60/90 s; 4 h	Remote SCADA / satellite practice

16. Phase 2 — model-based capabilities (off until their condition is met)

Capability	Condition that turns it on
Digital twin (physics + ML residual)	Physics cores per well class built; ≥ 30 days shadow; calibration monitored.
Fleet causal-graph reasoning + causal engine	Maintained field topology graph and ≥ 6 months interventional log.
Probabilistic world-model / joint inference	Twin + causal graph + inference engine all live.
Remaining-useful-life prediction	A labeled fleet failure history of meaningful size exists.
Stochastic forecasting / what-if	Twin live and calibrated.
Opportunity ranking	Forecasting live.
Fleet optimization	Forecasting + ranking live and one closed-loop setpoint case safety-signed.
Graduated closed-loop autonomy	Per-loop engineering safety case signed.
Online + structural learning, model governance	Learning infrastructure + watchdog live.
Calibration backbone, feature attribution, counterfactuals	Probabilistic models emitting real probabilities.
Real signing / provenance crypto	Identity + edge crypto stack deployed (unlocks full state suppression).

Capability	Condition that turns it on
Adversarial-ML robustness, threat-intel feed	ML layer exists; threat→action mapping defined.
Natural-language operator overrides	Parser specified; causal engine live. Until then, operators pick a reason from a fixed list.
Auto-drone dispatch	Drone system fielded; risk-score weights defined.

17. Site-specific inputs required before go-live

A trigger stays UNARMED + reason until its inputs are supplied.

- **Per-well:** `mawp_psi`, `maasp_psi`, `esp_flg_amps`, `esp_freq_setpoint_hz`, gas-handling limit, `wcut_limit_pct`, decline-forecast params, `hydrate_curve`/`wax_point_f`, `pipe_geometry`, `h2s_sour_baseline_ppm`, `esp_vib_ref`.
- **Per-channel:** `instrument_span`, `slew_max` (defaults in 3.1 used if absent).
- **Custody:** `custody_flow` (allocation meter) for executive theft detection; without it, T1-E is advisory-only.
- **Per-detector:** sand thresholds in the installed device's unit.
- **Per-site:** muster-zone map, FP cost table, alarm priority/response-time table, regional profile.
- **Go-live gate (H₂S):** any well with a sour baseline within 5 ppm of a safety band requires engineered, verified breathing-zone detection before it is cleared for live operation (Section 6.2).
- **Go-live gate (thermal wells):** any thermal-recovery well must have its temperature `instrument_span` raised to its true operating range (e.g. 0–600 °F) and steam-cycle-

aware SCP compensation configured before go-live (Sections 3.1, 6.3).

- **Per-unit OEM:** ESP ISO severity bands, motor-temp offsets, post-workover seed curves.
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18. Build order

1. Replay harness: validate triggers + state logic against labeled datasets (per-trigger, per-state precision/recall, lead-time, adversarial replays).
2. SAFE CORE at the edge: Layer-0, T1, gas response, ring escalation, modes, `degradation_score`.
3. Telemetry contract + reference data loaded; UNARMED tags working.
4. T2/T3 with flowing-only averages, the slow-trend water-cut arm, and clean-baseline ESP arms; coarse confidence; simple grouping; Governor by item-count.
5. KPIs + trust score on closed alerts; FP economics with the cost table, including the shut-in false-decline and sour-notify-rate measures.
6. Then, behind their conditions, the Section 16 capabilities — twin first, the rest layered on it.